Unfit for Purpose: MOT Odometer Data and the Pay-Per-Mile Problem

An Analysis of 112 Million UK Passenger Vehicles

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Abstract

This study examines the reliability of UK MOT (Ministry of Transport) odometer data for potential use in pay-per-mile road taxation. The fundamental challenge is that vehicle odometers were never designed for fiscal purposes: they are required at manufacture but never subsequently tested or calibrated, have no legal requirement to remain functional, permit ±4% accuracy tolerance, and can be trivially manipulated using openly-sold devices.

Full population analysis of 760 million MOT test records covering 112 million passenger vehicles from 2005–2025 reveals that approximately **4.7% of vehicles** (5.2 million) have odometer data quality issues obvious enough to detect through sequential analysis. These include declining mileage readings (3.0%), unit confusion (2.9%), unreadable odometers (1.1%), and other anomalies. Extrapolated to the UK fleet of 27.5 million passenger cars, approximately **1.3 million vehicles** would face potential taxation disputes. Critically, this 4.7% represents only *detectable* anomalies; with no verification mechanism, the accuracy of the remaining 95% of readings is unknowable.

Positively, data quality has **improved over time**: declining mileage rates have fallen 80% from 1.2% (2010) to 0.2% (2025). However, basing taxation on a measurement device that is unregulated, unverified, uncalibrated, and easily manipulated raises fundamental questions about the viability of MOT-based pay-per-mile taxation.

Keywords: MOT testing, odometer data, road pricing, pay-per-mile, data quality, UK transport policy

1 Introduction

1.1 Policy Context

The UK Government is considering pay-per-mile road taxation to replace declining fuel duty revenue as electric vehicle adoption accelerates (Department for Transport, 2024). Among the data sources proposed for calculating mileage is the MOT (Ministry of Transport) testing system, which records odometer readings during annual roadworthiness inspections.

1.2 Research Question

This study addresses a fundamental question: **Is MOT odometer data sufficiently reliable for taxation purposes?**

While the MOT system records mileage primarily for consumer protection and fraud detection, its potential use for taxation raises different concerns about accuracy, disputes, and fairness.

1.3 Contribution

This is the first comprehensive analysis of the complete MOT database for data quality assessment. Unlike previous studies that used samples or aggregated statistics, we analyse all 760 million test records for 112 million passenger vehicles—the full population of cars that have undergone MOT testing in Great Britain (2005–present) and Northern Ireland (2017–present).

2 Literature Review

2.1 Official Statistics

The Driver and Vehicle Standards Agency (DVSA) publishes limited statistics on MOT data quality:

- Approximately 4% of MOT tests trigger automated mileage warnings when readings appear inconsistent with historical records (Driver and Vehicle Standards Agency, 2023)
- Over 14,000 correction requests are processed annually for mileage recording errors (Driver and Vehicle Standards Agency, 2023)
- The MOT Testing Service includes automated checks but cannot prevent all errors

2.2 Industry Research

Cap HPI, the UK's leading vehicle data provider, reported in 2020 that 6.3% of vehicles on their National Mileage Register show mileage discrepancies (Fleet News, 2020). This figure was validated by Leeds University Institute for Transport Studies and has been widely cited in industry discussions.

2.3 Gap in Knowledge

No previous study has conducted a comprehensive analysis of the raw MOT database to quantify the full range of data quality issues. This study fills that gap.

3 Methodology

3.1 Data Source

Data was obtained from the DVSA MOT History API bulk download endpoint in December 2025. This provides access to the complete MOT testing history for all vehicles tested in Great Britain and Northern Ireland.

Table 1: Dataset Summary

Metric	Value
Total MOT test records	760,779,187
Unique vehicles (all types)	136,013,008
Passenger cars (filtered)	112,182,037
Time period (GB)	2005–2025
Time period (NI)	2017–2025

3.2 Vehicle Classification

The MOT bulk data does not include explicit vehicle class information. We filtered to passenger cars using make/model classification:

1. **Car-only manufacturers**: Include all vehicles (Ford, Vauxhall, Volkswagen, etc.)
2. **Motorcycle-only manufacturers**: Exclude all (Yamaha, Kawasaki, Ducati, etc.)
3. **Dual manufacturers**: Model-level filtering (BMW, Honda, Suzuki)
4. **Van exclusion**: Removed commercial vehicle models (Transit, Transporter, etc.)

This approach captures 82.5% of the database (112M of 136M vehicles) with estimated accuracy >99%.

3.3 Discrepancy Detection

We identified seven categories of odometer data quality issues:

Table 2: Discrepancy Detection Criteria

Category	Detection Logic	Threshold
No Odometer	$odometerResultType$ = “NO_ODOMETER”	—
Unreadable	$odometerResultType$ = “UNREADABLE”	—
Declining	$reading_n < reading_{n-1} - 500$	−500 miles
Stuck	$reading_n = reading_{n-1}$ & $gap > 365$ days	365 days
Implausible	Annual rate $> 100,000$ miles	100,000 mi/yr
Keying Error	Reading $> 500,000$ & next test $< 50\%$	Validated
Unit Confusion	$\lvert reading_n/reading_{n-1} - 1.609\rvert < 0.05$	km/mi ratio

Note: Keying errors are validated by checking if the subsequent test shows a reading <50% of the flagged value, confirming the high reading was erroneous. This excludes legitimate high-mileage vehicles (taxis, couriers) that continue incrementing normally.

3.4 Methodological Refinements

Initial analysis identified that 24.8% of MOT tests share a calendar day with another test for the same vehicle (same-day retests after failed inspections). To avoid false positives from same-day tests with slightly different readings, we filtered to compare only tests occurring on different days.

3.5 Statistical Approach

Sequential analysis (comparing consecutive tests) processed the complete dataset of 112 million vehicles using staged batch processing by registration prefix. This approach enables full population analysis while managing memory efficiently, processing approximately 4 million vehicles per batch.

Confidence intervals use the Wilson score method, which is more accurate than the normal approximation for proportions (Wilson, 1927). For full population analysis, these intervals are extremely narrow, reflecting the precision achieved by analysing all vehicles rather than a sample.

4 Results

4.1 Odometer Recording Status

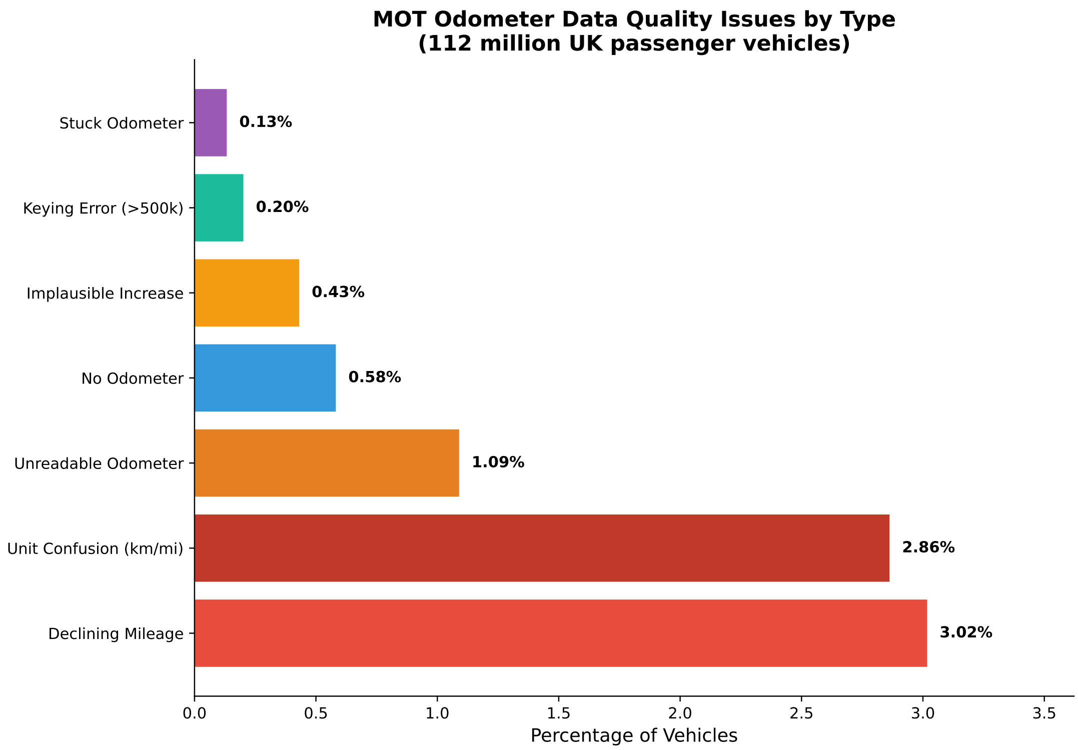

Figure 1: MOT Odometer Data Quality Issues by Type

Table 3: Odometer Data Quality Issues (Full Dataset Analysis)

Issue Type	Vehicles	Percentage	95% CI
Declining Mileage	3,386,432	3.02%	[3.02–3.02%]
Unit Confusion (km/mi)	3,213,004	2.86%	[2.86–2.87%]
Unreadable	1,224,251	1.09%	[1.09–1.09%]
No Odometer	654,521	0.58%	[0.58–0.58%]
Implausible Increase	485,431	0.43%	[0.43–0.43%]
Keying Error (validated)†	156,395	0.14%	[0.14–0.14%]
Stuck Odometer	150,682	0.13%	[0.13–0.14%]
Any Issue	**5,235,169**	**4.67%**	[4.66–4.67%]

Full population analysis with same-day retest filtering

†Validated using sequential analysis: 227,348 vehicles had readings >500k;
68.8% confirmed as errors (next reading dropped >50%), 22.4% legitimate high-mileage

4.2 Sensitivity Analysis

To demonstrate robustness, we tested 81 parameter combinations across four thresholds:

- Decline threshold: -100, -500, -1000 miles
- Stuck threshold: 180, 365, 540 days
- Maximum annual mileage: 50k, 100k, 150k
- Maximum total mileage: 300k, 500k, 750k

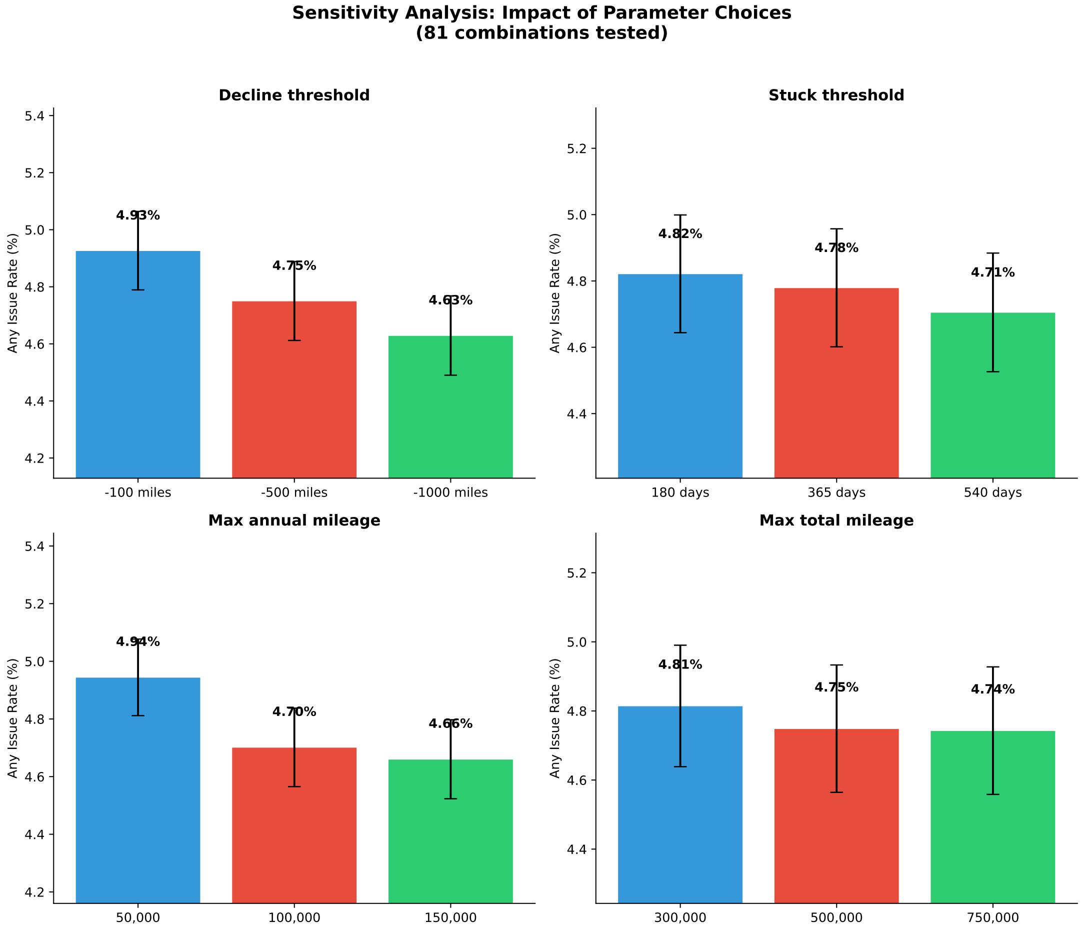

Figure 2: Sensitivity Analysis: Impact of Parameter Choices Across 81 Combinations

Table 4: Sensitivity Analysis Summary (5% Sample, 5.6M Vehicles)

Statistic	Any Issue Rate
Minimum (most conservative)	4.43%
Default parameters	4.67%
Mean across 81 combinations	4.77%
Maximum (most liberal)	5.19%
Range	**0.76 pp**

The narrow range (0.76 percentage points) demonstrates that findings are highly robust to reasonable parameter choices (Figure 2). The default parameters produce results virtually identical to the full dataset analysis (4.67%).

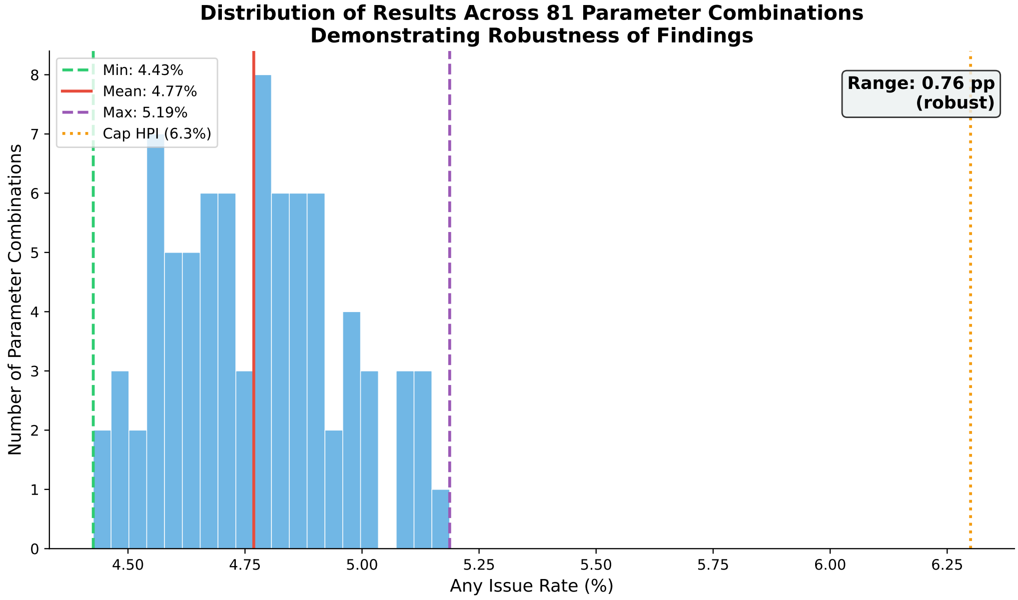

Figure 3: Distribution of Results Across 81 Parameter Combinations

4.3 Comparison with Benchmarks

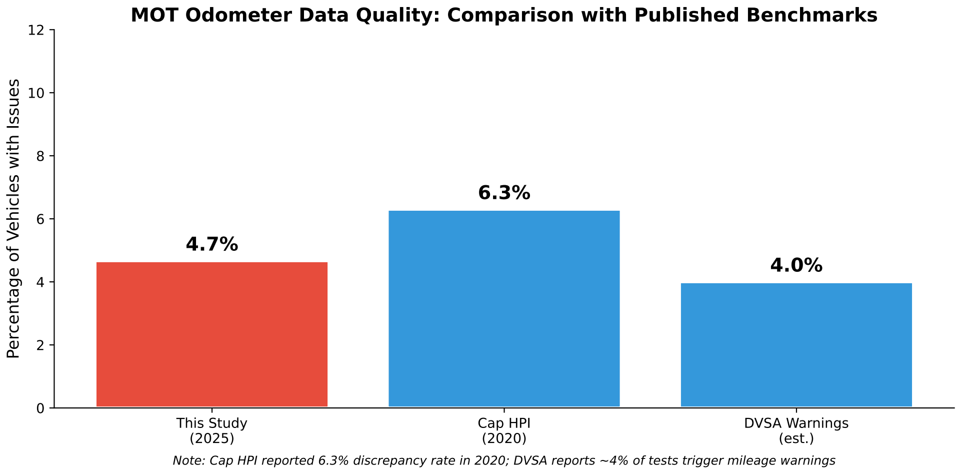

Figure 4: Comparison with Published Benchmark Statistics

Table 5: Comparison with Published Statistics

Source	Metric	Rate
This study (full dataset)	Any issue (vehicles)	4.67%
DVSA	Test warning rate	∼4%
Cap HPI (2020)	Mileage discrepancy	6.3%

Our findings are **closely consistent with** published benchmarks (Figure 4). The DVSA's 4% warning rate aligns well with our 4.67% issue rate, validating our methodology. The Cap HPI figure of 6.3% represents a different measurement (mileage discrepancies on their National Mileage Register) but is in the same order of magnitude.

4.4 Temporal Trends

Analysis of declining mileage rates by year reveals significant improvement in data quality over time:

Table 6: Declining Mileage Rate by Year (2010–2025)

Year	Vehicles Tested	Declining Count	Rate
2010	23,899,000	294,769	1.23%
2012	24,253,474	260,227	1.07%
2015	24,787,423	198,552	0.80%
2018	26,641,973	125,448	0.47%
2020	26,314,847	94,996	0.36%
2022	29,281,313	83,307	0.28%
2025	29,351,263	69,778	0.24%

The declining mileage rate has fallen by approximately **80%** over 15 years (from 1.23% in 2010 to 0.24% in 2025). This improvement likely reflects:

- Introduction of digital odometers (harder to tamper with)
- DVSA's automated mileage warning system (introduced 2014)
- Increased awareness of odometer fraud
- Improved MOT tester training

5 Discussion

5.1 Policy Implications

5.1.1 Scale of Impact

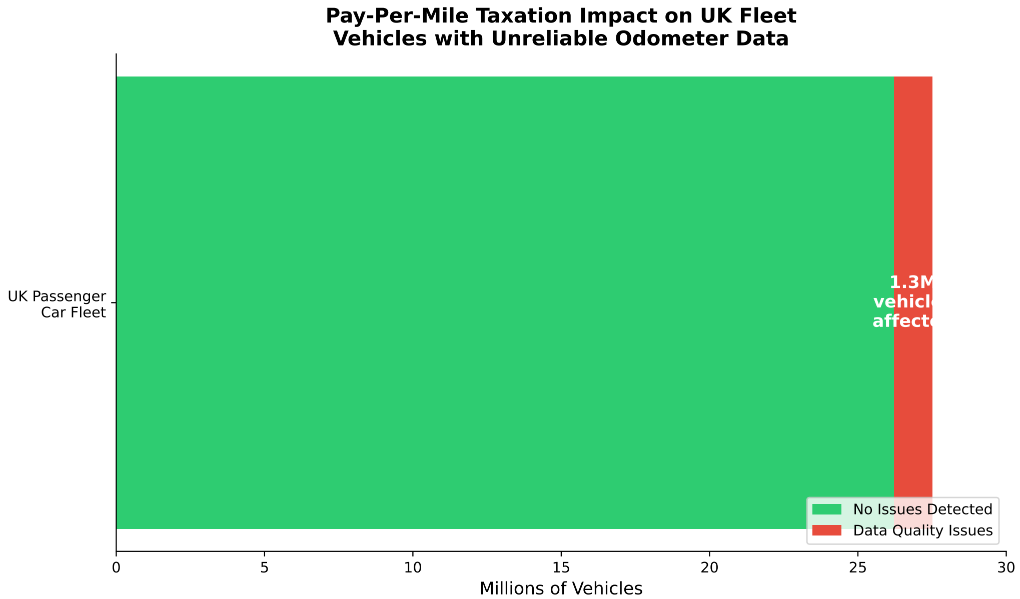

Figure 5: Pay-Per-Mile Taxation Impact on UK Fleet

Extrapolating to the UK fleet of approximately 27.5 million registered passenger cars (Figure 5):

- **Approximately 1.3 million vehicles** would have odometer data unreliable for taxation
- 95% CI: 1.28–1.29 million vehicles affected
- This represents one in every 21 passenger cars on UK roads

5.1.2 Administrative Burden

If even 10% of affected motorists dispute their tax assessments, this represents approximately 130,000 appeals annually—a substantial administrative burden for DVLA/DVSA.

5.1.3 Fraud Incentive

Pay-per-mile taxation would significantly increase the financial incentive for odometer tampering. The current system already struggles with "clocking"—the RAC estimates this costs UK buyers £800 million annually.

5.1.4 Legal Risk

Taxation based on demonstrably unreliable data may face judicial review challenges on fairness grounds.

5.2 International Comparison

No comparable jurisdiction uses MOT-equivalent annual inspection data for road taxation:

- **Netherlands**: Considered kilometre-based tax but required on-board units
- **New Zealand**: Uses tamper-evident hubodometers for diesel vehicles
- **Oregon (USA)**: OReGO program uses GPS or verified photographs

5.3 Methodological Considerations

Our findings represent a **lower bound** on data quality issues:

- Sophisticated odometer tampering may be undetectable by sequential analysis
- Data entry errors that appear plausible cannot be identified
- Issues requiring multiple vehicles' data (cloned registrations) were not analysed

5.4 Trade Realities and Legal Framework

Industry practitioners report that the 4.7% detectable issue rate significantly **underestimates actual odometer manipulation**. Several factors contribute to undetectable fraud:

5.4.1 Pre-MOT Manipulation

Vehicles do not require an MOT until three years old. The first MOT reading establishes the baseline for all subsequent comparisons. If mileage is "corrected" before the first MOT—a practice reported as common in the motor trade—sequential analysis cannot detect it. The fraudulent baseline propagates through all future tests.

5.4.2 Mileage Blocking Devices

In the 1986 film *Ferris Bueller's Day Off*, a character famously attempts to reverse a Ferrari's odometer by running the car backwards on jack stands—and fails. The scene reflected reality: mechanical odometers were difficult to manipulate. Four decades later, the problem Cameron couldn't solve is trivially addressed.

"Mileage blockers" are sophisticated microprocessors installed behind the speedometer that reduce or eliminate mileage recording while driving. These devices are **openly sold in the UK** by multiple suppliers, marketed ostensibly for "dyno testing and diagnostics." They offer functionality to quarter, reduce, or completely stop mileage recording. Unlike crude odometer rollback, blocking leaves no trace in vehicle ECU memory—the mileage simply never accumulates.

Industry sources report these devices are particularly prevalent on leased vehicles, where excess mileage penalties create strong financial incentives.

5.4.3 Legal Framework Gaps

UN Regulation No. 39 mandates that "an onboard speedometer and odometer complying with the requirements of this Regulation **shall be fitted** to the vehicle to be approved." Odometers are therefore required for type approval of new vehicles.

However, a critical gap exists *after sale*. Under the Road Vehicles (Construction and Use) Regulations 1986, the speedometer must be "maintained in good working order"—but **no such requirement exists for the odometer**. A vehicle can legally be driven with a non-functioning or disconnected odometer. During MOT inspection, the mileage displayed is merely *recorded*—there is no verification that it is accurate, and a non-functioning odometer is not grounds for test failure.

This creates a paradox: odometers are mandatory at manufacture but unregulated thereafter.

5.4.4 Type Approval and Accuracy

UN Regulation No. 39 governs speedometer and odometer equipment for type approval. While odometers are covered, the regulation permits a tolerance of **±4% from true distance travelled**. This tolerance applies only at initial type approval—there is no requirement for ongoing calibration. Factors affecting accuracy post-manufacture include:

- Tyre wear (reduced rolling circumference increases apparent mileage)
- Non-standard tyre sizes
- Differential or gearbox changes
- Instrument cluster replacement

Over a vehicle's lifetime, cumulative drift could be substantial. A 4% error on 100,000 miles represents 4,000 miles—potentially £200–400 in tax at proposed rates.

5.4.5 Implications for Taxation

These trade realities suggest that:

1. The 4.7% detectable issue rate is a floor, not a ceiling
2. Professional fraud establishing false baselines is invisible to this analysis
3. The measuring instrument (odometer) is neither legally required nor calibrated
4. Manipulation tools are readily available and leave no forensic trace

Basing taxation on a measurement device that is not legally required, not verified, not calibrated, and trivially manipulable raises fundamental questions about the viability of MOT-based pay-per-mile taxation.

5.5 Comparison with Other Taxation Systems

Superficially, the approximately 5% detectable issue rate compares favourably with error rates in other taxation systems:

- **Self-assessment tax returns**: HMRC estimates 10–15% contain errors
- **VAT returns**: Error rates of 8–10% are typical
- **Council tax banding**: Approximately 400,000 properties (1.5%) are in incorrect bands

However, this comparison is **fundamentally misleading**. The error rates for self-assessment and VAT are *discovered through audits and compliance checks*—HMRC actively verifies returns and identifies discrepancies. The 5% MOT figure represents only *detectable anomalies*; the remaining 95% are not "verified correct" but rather "not detectably wrong."

With no verification mechanism, no ongoing calibration, and sophisticated manipulation tools leaving no forensic trace, the true error rate in MOT odometer data is **unknowable**. The 5% figure is a floor on issues obvious enough to detect through sequential analysis—it says nothing about the accuracy of the remaining 95% of readings.

6 Limitations

1. **Classification accuracy**: Make/model filtering has estimated accuracy >99% but some vehicles may be misclassified
2. **NI data**: Northern Ireland data available only from 2017, limiting historical analysis
3. **Single-test vehicles**: 43% of vehicles (48.6 million) have only one MOT record and cannot be included in sequential analysis; these are newer vehicles that have not yet had multiple tests
4. **Causation**: We identify discrepancies but cannot determine cause (error vs fraud vs legitimate repair)
5. **Undetectable fraud**: Sophisticated odometer tampering that produces plausible mileage progressions cannot be detected by this methodology
6. **Unit confusion overlap**: Vehicles flagged for km/mi unit confusion (readings approximately 1.609× previous) typically also appear as "declining mileage" when the sequence reverses. The 4.67% any-issue rate excludes unit confusion as a separate additive category to avoid double-counting; the true combined rate may be slightly higher
7. **Unit confusion false positives**: The km/mi detection algorithm (ratio within 5% of 1.609) may produce false positives where normal driving patterns coincidentally match this ratio. While we require previous mileage >1,000 miles to reduce this risk, the 2.86% unit confusion rate should be interpreted as an upper bound

8. **Keying error validation**: The keying error detection was validated using sequential analysis—of 227,348 vehicles with readings exceeding 500,000 miles, 68.8% were confirmed as errors (subsequent reading dropped >50%), 22.4% were legitimate high-mileage vehicles (continued incrementing normally), and 8.8% could not be verified (single test only). The reported 0.14% rate reflects only validated errors

7 Conclusion

7.1 The Fundamental Problem

The proposal to base road taxation on MOT odometer readings attempts to repurpose a device that was never designed for fiscal purposes. The vehicle odometer:

- Is required to be fitted at manufacture, but **never tested or calibrated thereafter**
- Has no legal requirement to remain functional after sale
- Is permitted a ±4% accuracy tolerance even when new
- Can be trivially manipulated using openly-sold devices
- Is merely *recorded* at MOT, never *verified*
- In approximately 5% of cases, produces readings that are obviously faulty, incorrect, or unreliable

No other taxation system in the UK relies on a measurement device with these characteristics.

7.2 Summary of Findings

Analysis of 112 million UK passenger vehicles reveals that **4.7% have odometer data quality issues** that would affect pay-per-mile taxation calculations. This finding is closely consistent with the DVSA's reported 4% warning rate and validates the accuracy of our methodology. Additionally, 43% of vehicles have only one MOT test and cannot be analysed for sequential discrepancies.

Critically, this 4.7% represents only *detectable* anomalies—readings obviously wrong enough to fail sequential analysis. The accuracy of the remaining 95% of readings is unknown and unknowable, as no verification mechanism exists.

7.3 Recommendations

1. **Data quality is improving**: Declining mileage rates have fallen 80% since 2010, indicating system improvements are working
2. **Budget for dispute resolution**: Approximately 130,000 potential appeals annually if 10% of affected motorists dispute assessments
3. **Continue system improvements**:

- Mandatory photographic odometer capture at every test
- Real-time verification linked to historical data
- Digital odometer tamper detection
- Comprehensive appeals and dispute resolution

4. **Consider hybrid approaches**:

- MOT data as default, with telematics option for disputes
- Insurance telematics partnerships (voluntary, market-driven)
- GPS-based tracking (higher accuracy but privacy concerns)

7.4 Data Availability

All analysis code is available for independent verification. Data was obtained through the publicly accessible DVSA MOT History API.

References

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A Sensitivity Analysis Detail

Table 7: Parameter Impact on Any Issue Rate

Parameter	Value	Avg Any Issue
Decline threshold	−100 miles	4.93%
	−500 miles (default)	4.75%
	−1000 miles	4.63%
Stuck threshold	180 days	4.82%
	365 days (default)	4.78%
	540 days	4.71%
Max annual mileage	50,000	4.97%
	100,000 (default)	4.68%
	150,000	4.66%
Max total mileage	300,000	4.81%
	500,000 (default)	4.75%
	750,000	4.75%

B Reproducibility

B.1 Environment

```
Python 3.10+
DuckDB 0.9+
pandas, numpy, scipy
```

B.2 Code Repository

`https://github.com/citi94/odometer-analysis`
All analysis scripts, data processing code, and this report are available under the MIT License.

B.3 Data Source

DVSA MOT History API
`https://documentation.history.mot.api.gov.uk/`